CS & IT ENGINEERING

Operating System

Virtual Memory

Lecture - 4

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Recap of Previous Lecture

Topic Page Replacement

Topic FIFO, Optimal, LRU Policies

Topics to be Covered

Topic LIFO, LFU, MFU, MRU Policies

Topic Page Replacement Algorithms

Topic : Question

- Number of frames = 4
- Pure demand paging used
- Page reference sequence: 5, 7, 0, 1, 7, 6, 7, 2, 1, 6, 7, 6, 1
- Number of page faults for FIFO, optimal and LRU policies?

7 6 6

Consider the following page references:

2, 3, 4, 5, 6, 4, 5, 2, 7, 8, 9, 8, 9, 8, 9, 1, 6, 5, 6, 5, 3

Using optimal policy and 4 frames. Memory access time is 2ms without page fault and 40ms with page fault. The $\underbrace{\text{effective}}_{\downarrow \text{avg.}}$ memory access time for servicing the above page requests is ____ ms?

Page fault rate $= \frac{10}{21}$

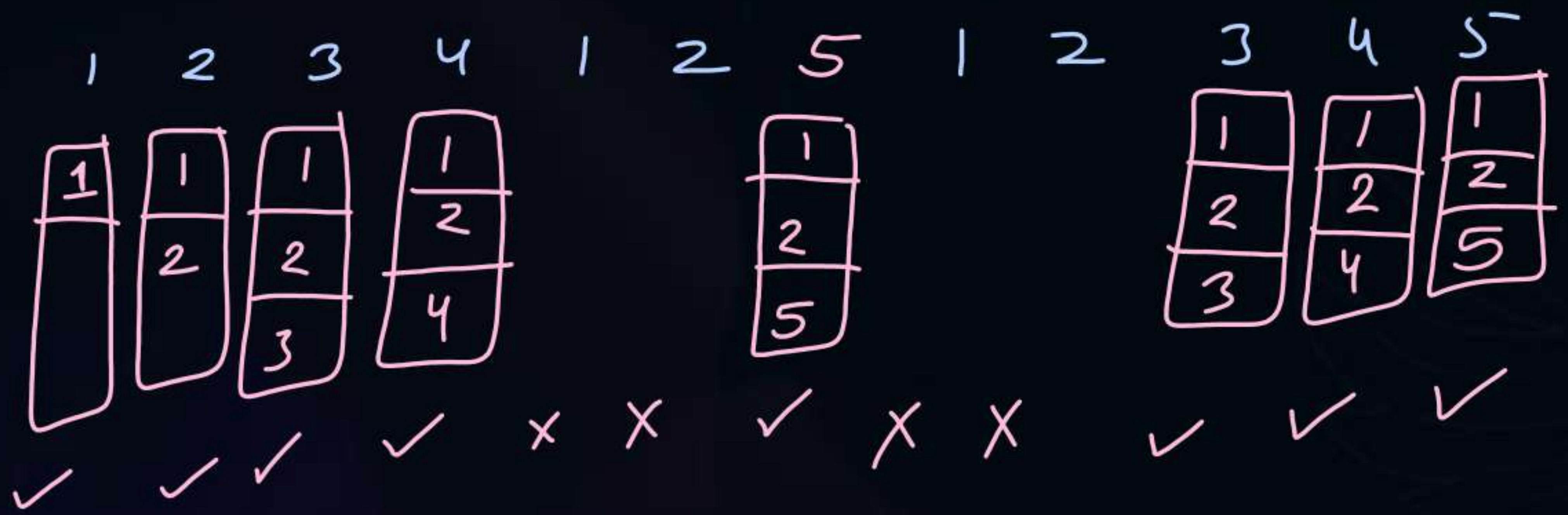
$$EMAT = \frac{10}{21} * 40\,ms + \frac{11}{21} * 2\,ms$$

$$= 20.1\ ms$$

Topic : Last In First Out (LIFO)

Assume:

- Number of frames = 3 (All empty initially)
- Page reference sequence: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5



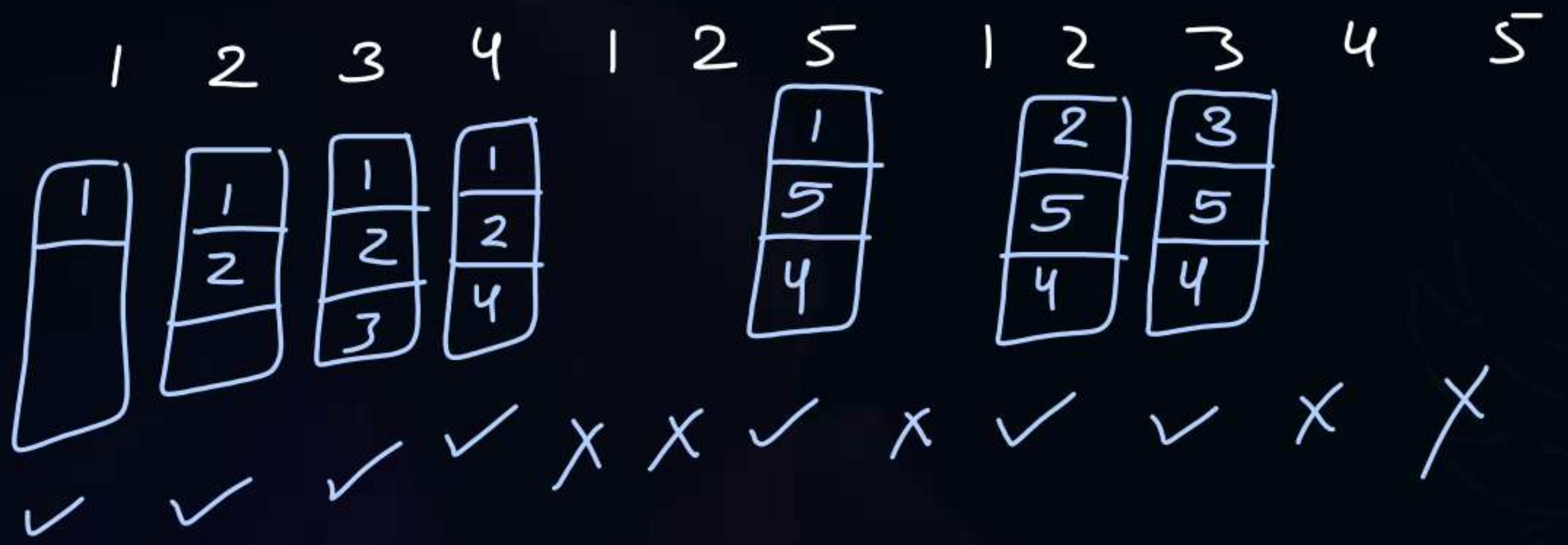
no. of page faults = 8

Topic : Most Recently Used (MRU)

replace the page which has been used most recently

Assume:

- Number of frames = 3 (All empty initially)
- Page reference sequence: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5



no. of faults = 7

Topic : Counting Algorithms

- Counting algorithms look at the number of occurrences of a particular page and use this as the criterion for replacement.

- Such counting algorithms includes:
  - LFU (Least Frequently Used)
  - MFU (Most Frequently Used)

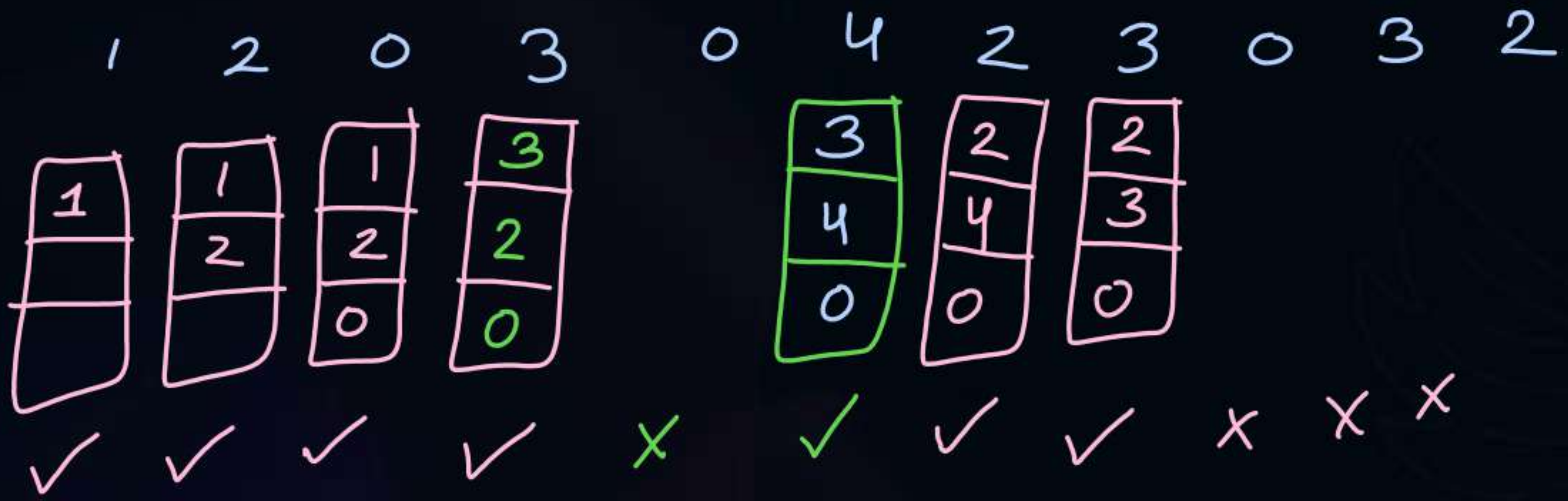
Topic : Least Frequently Used (LFU)

tie breaker ⇒ FIFO

replace a page which has been used min. no. of times.

Assume:

- Number of frames = 3 (All empty initially)
- Page reference sequence: 1 2 0 3 0 4 2 3 0 3 2



page	frequency
1	1
2	~~1~~ ~~2~~ 3
3	~~1~~ ~~2~~ 3
0	~~1~~ ~~2~~ 3
4	1

no. of faults = 7

Topic : Most Frequently Used (MFU)

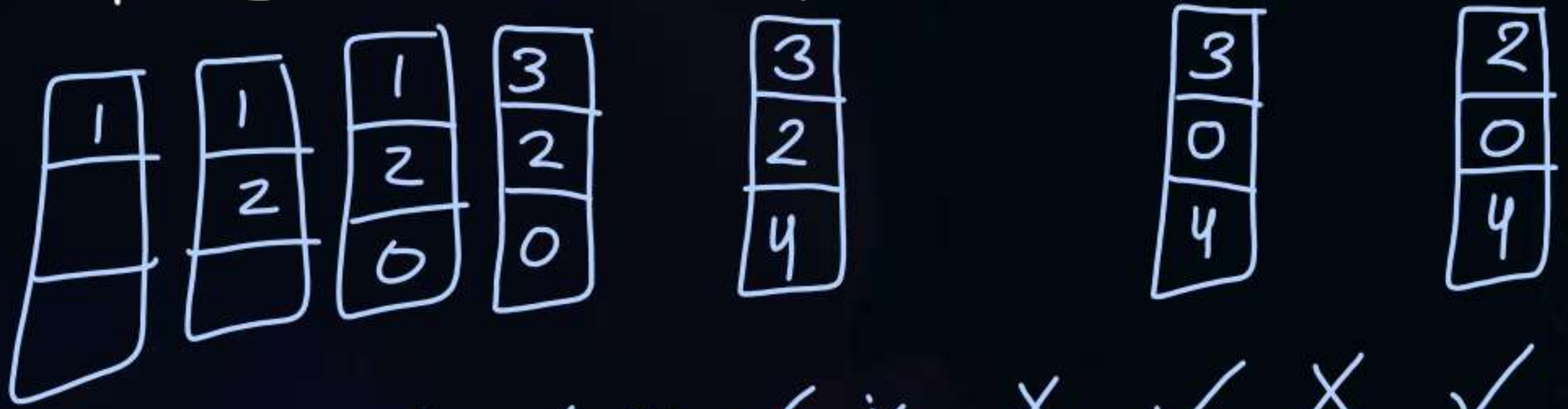
tie breaker ⇒ FIFO

replace the page which has been used max. no. of times

Assume:

- Number of frames = 3 (All empty initially)
- Page reference sequence: 1 2 0 3 0 4 2 3 0 3 2

1 2 0 3 0 4 2 3 0 3 2



✓ ✓ ✓ ✓ X ✓ X X ✓ X ✓

no. of page faults = 7

Page	frequency
0	~~1~~ ~~2~~ 3
1	1
2	~~1~~ ~~2~~ 3
3	~~1~~ ~~2~~ 3
4	1

Topic : Question

$31 - 30 = \underline{\underline{1}}$ Ans

[GATE-2016]

#Q. Consider a computer system with ten physical page frames. The system is provided with an access sequence $a_1, a_2, \ldots, a_{20}, a_1, a_2, \ldots, a_{20}$ where each ai is page number. The difference in the number of page faults between the last-in-first-out page replacement policy and the optimal page replacement policy is ________

LIFO :- ✓a_1 ✓a_2 ✓a_3 .... ✓a_{10} ✓a_{11} ✓a_{12} ✓a_{13} ✓a_{14} ... ✓a_{20} ×a_1 ×a_2 .. ×a_9 ✓a_{10} ....✓ a_{20}

a_{10}	a_{11}	a_{12}	a_{20}	a_{10}	a_{20}
a_1	a_1	a_1	a_1	a_1	a_1
a_2	⋮	⋮	⋮	⋮	⋮
⋮					
a_9	a_9	a_9	a_9	a_9	a_9
a_{10}	a_{11}	a_{12}	a_{20}	a_{10}	a_{20}

faults = 31

optimal :-

a_1	a_2	$\cdots$	a_9	a_{10}	a_{11}	a_{12}	$\cdots$	a_{20}	a_1	a_2	$\cdots$	a_9	a_{10}	a_{11}	a_{12}	$\cdots$	a_{20}
✓	✓		✓	✓	✓	✓		✓	×	×		×	✓	✓	✓	✓	×
				a_1	a_1	a_1	$\cdots$	a_1					a_{10}				
				a_2	$\vdots$	$\vdots$		$\vdots$					a_2				
				$\vdots$									$\vdots$				
				a_9	a_9	a_9		a_9									
				a_{10}	a_{11}	a_{12}		a_{20}					a_{20}				

faults = 30

#Q. A main memory can hold 3 page frames and initially all of them are vacant. Consider the following stream of page requests :

2, 3, 2, 4, 6, 2, 5, 6, 1, 4, 6

If the stream uses FIFO replacement policy, the hit ratio h will be ?

✓	✓	×	✓	✓	✓	✓	×	✓	✓	✓
2	2		2	6	6	6		1	1	1
	3		3	3	2	2		2	4	4
			4	4	4	5		5	5	6

A. 11/3

B. 1/11

C. 3/11

D. ✓ 2/11

Topic : Question

#Q. A virtual memory system has only 2-page frames which are empty initially. Using demand paging the following sequence of page reference is passed through this system.

9, 8, 7, 8, 7, 9, 7, 9, 8, 9

Minimum possible number of page faults ? ⇒ 5 Ans.

optimal policy

✓	✓	✓	×	×	✓	×	×	✓	×
9	9	7			7			8	
	8	8			9			9	

Topic : Question

[GATE-2007]

#Q. The address sequence generated by tracing a particular program executing in a pure demand paging system with 100 bytes per page is
0100, 0200, 0430, 0499, 0510, 0530, 0560, 0120, 0220, 0240, 0260, 0320, 0410.
Suppose that the memory can store only one page and if x is the address which causes a page fault then the bytes from addresses x to x + 99 are loaded on to the memory.
How many page faults will occur?

A 0

B 4

C 7

D 8

frame ⇒ ~~0100 — 0199~~

~~0200 — 0299~~

~~0430 — 0529~~

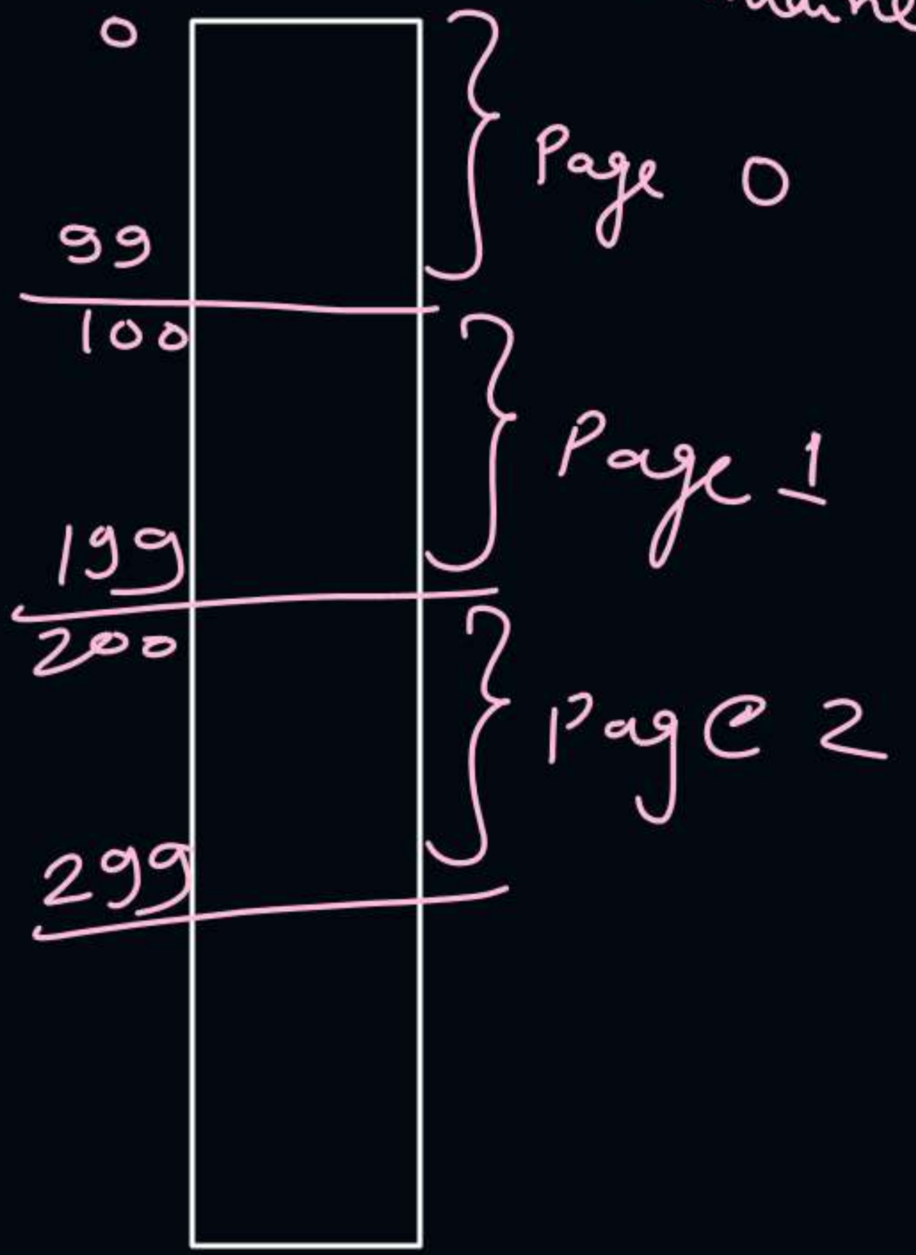
~~0530 — 0629~~

~~0120 — 0219~~

~~0220 — 0319~~

0320 — 0419

if x to x+99 line will not be mentained



Topic : Making Page Reference Sequence

if L.A. is given

L.A.
- In binary ⇒
- In hexadecimal
- In decimal

L.A.

p	d

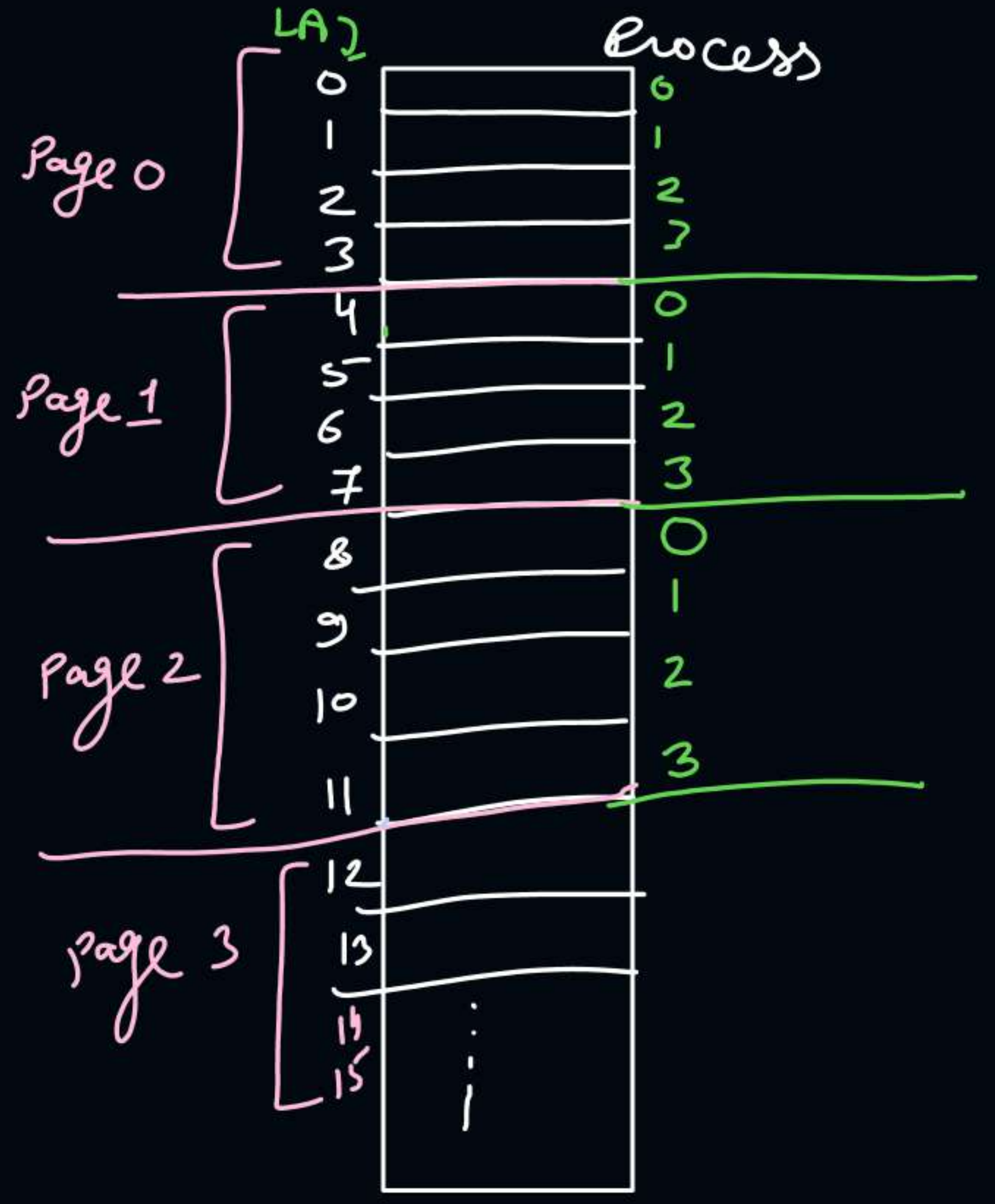
P.A.

f	d

$$\text{Page no.} = \left\lfloor \frac{L.A.}{\text{Page size}} \right\rfloor$$

$$\text{offset} = L.A. \ \% \ \text{Page size}$$

ex:- assume page size = 4 bytes



if LA = 11

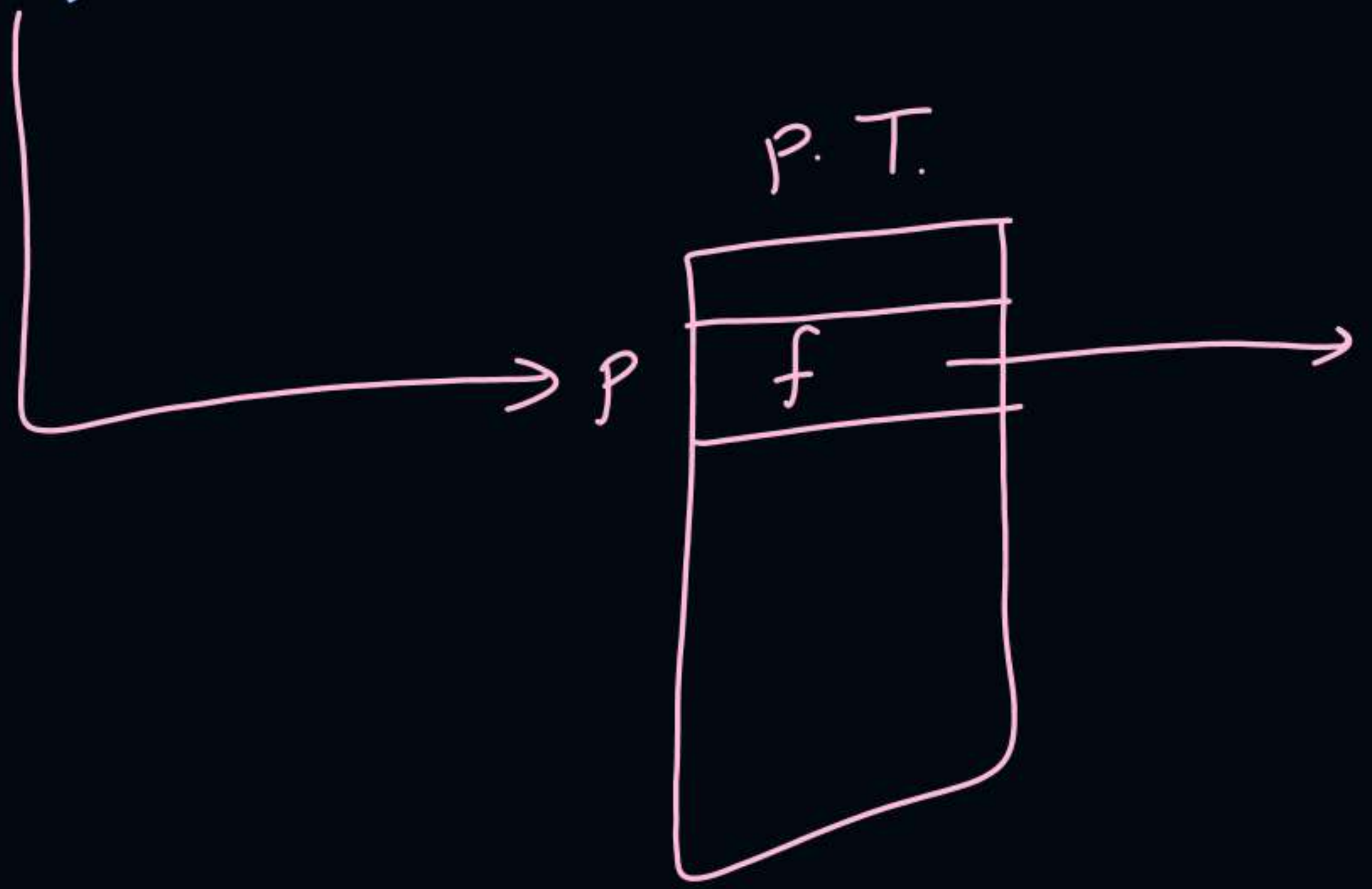
$$P = \left\lfloor \frac{11}{4} \right\rfloor = 2$$

$$d = 11 \% 4 = 3$$

L.A. in decimal

$\Downarrow$

p, d calculated



P.A. $\Rightarrow (f * \text{page size}) + d$

2 mins Summary

Topic LIFO, LFU, MFU, MRU Policies

Topic Page Replacement Algorithms

Happy Learning

THANK - YOU